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MultiModal-GPT: A Vision and Language Model for Dialogue with Humans

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Abstract

We present a vision and language model named MultiModal-GPT to conduct multi-round dialogue with humans. MultiModal-GPT is capable of following diverse instructions, such as generating detailed captions, counting specific objects, and addressing general inquiries posed by users. The model is efficiently fine-tuned from OpenFlamingo, with Low-rank Adapter (LoRA) incorporated in both the gated-cross-attention and self-attention components of the language model. Our approach involves constructing instruction templates that incorporate vision and language data for multi-modality instruction tuning, enabling the model to comprehend and adhere to human directives. We observe that the quality of training data is crucial for effective dialogue performance, as a limited dataset with short responses may cause the model to generate brief replies to any instruction. To further enhance MultiModal-GPT's conversational abilities, we employ language-only instruction-following data for joint training alongside visual-language instructions. Utilizing the *same* instruction template for both types of data results in a significant improvement in dialogue performance. Our experiments demonstrate MultiModal-GPT's proficiency in maintaining continuous dialogues with humans. The code, dataset, and demo can be found at https://github.com/open-mmlab/Multimodal-GPT.

1 Introduction

Humans interact with the world through multiple channels, including vision and language, each of which has a unique advantage in representing and conveying certain concepts of the world, thus contributing to a better understanding of the world. A central objective of artificial intelligence research is to create a versatile assistant capable of effectively following multimodal vision-and-language instructions that align with human intentions, in order to accomplish a diverse array of real-world tasks.

Recently, GPT-4 [11] has demonstrated remarkable proficiency in multi-modal dialogues with humans. Although GPT-4's [11] exceptional capabilities have been observed, the mechanisms underpinning its outstanding performance remain elusive. Studies such as Mini-GPT4 [18] and LLaVA [8] have sought to replicate this performance by aligning visual representations with the input space of LLM, subsequently utilizing the original self-attention in the LLM to process visual information. However, incorporating such models with detailed or spatiotemporal visual information can be computationally intensive due to the potentially large number of image tokens. Furthermore, both models employ

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vicuna [2], an open-source chatbot refined through fine-tuning LLaMA [16] on user-generated conversations from ChatGPT, which omits the language instruction tuning phase in their research.

To address these challenges, we build upon the open-source Flamingo framework [1], a multimodal pre-trained model that deploys a perceiver resampler to efficiently extract visual information from the vision encoder, while also employing gated cross-attention layers for image-text interactions. This model has been pre-trained on an extensive dataset of image-text pairs, showcasing robust few-shot visual comprehension capabilities. Nevertheless, it lacks the capacity to engage in zero-shot multi-turn image-text dialogues. As a result, our goal is to fine-tune OpenFlamingo using comprehensive datasets of image and text instructions, enabling the model to conduct conversations that more closely align with human preferences. By capitalizing on OpenFlamingo's foundational strengths, we aspire to narrow the performance gap between the model's existing capabilities and the desired outcome of more accurate, human-like interactions in multimodal dialogues. We have dubbed our multimodal chatbot MultiModal-GPT.

We also use a unified instruction template for both language and visual instruction data during model training. We first construct instruction templates with vision and language data to train the MultiModal-GPT. We find the training data is vital with respect to the performance of the MultiModal-GPT. Some datasets, such as VQA v2.0 [3], OKVQA [9], GQA [5], CLEVR [6] and NLVR [15] datasets, will degrade the dialogue performance of the MultiModal-GPT, since the response in these datasets is restricted to one or two words (e.g., yes/no). Consequently, when these datasets are incorporated into the training process, the model exhibits a tendency to generate answers comprising merely one or two words. This brevity is not conducive to user-friendliness.

To further enhance the ability of MultiModal-GPT to chat with people, we also collect language data and define a unified instruction template to jointly train the MultiModal-GPT. The joint training of language-only instructions and visual and language instructions effectively improves the performance of the model. We show various demos to show the ability of continuous dialogue of MultiModal-GPT with humans.

2 Unified Instruction Template

We propose a unified template for the integration of unimodal linguistic data and multimodal vision-and-language data, with the objective of effectively training the MultiModal-GPT model in a synergistic manner. This unified approach aims to enhance the model's performance across diverse tasks by leveraging the complementary strengths of both data modalities and fostering a more profound understanding of the underlying concepts.

2.1 Language-only Instruction Template

<BOS> Below is an instruction that describes a task. Write a response that appropriately completes the request
Instruction: {instruction}
Input: {input}
Response: {response} <EOS>

Table 1: The input sequence of language data used to train the model. The {instruction}, {input} and {response} are texts from the source data. Only the {response} part and <EOS> token will be calculated loss.

We employ the Dolly 15k and Alpaca GPT4 datasets [12] as resources for assessing language-only instruction-following capabilities. These datasets have been specifically designed to improve the performance of language models in executing instruction-based tasks. To ensure consistent instruction-following format, we utilize the prompt template presented in Table 1 for structuring the dataset input.

<BOS> Below is an instruction that describes a task. Write a response that appropriately completes the request
Image: <image_token>
Instruction: {question}
Response: {response}<EOS>
Instruction: {question}
Response: {response} <EOS>

Table 2: The input sequence of vision and language data used to train the model. The {question} and {response} are texts from the source data. <image_token> is a token denoting the existence of image. Note that there are multi-round dialogues if the dataset has. Only the {response} part and <EOS> token will be calculated loss.

- Can you describe the image?
- Could you provide a description of the image?
- What do you see in this image?
- Share your thoughts on the content of the image.
- Please narrate what's happening in the picture.
- Can you give a brief explanation of the image?
- Describe the main elements and details present in the image.
- In your own words, what is depicted in the image?
- How would you describe the image's content in a caption?
- Can you suggest an insightful caption that highlights the underlying message of the image?

Table 3: The list of instructions for image caption.

2.2 Vision and Language Instruction Template

We utilize a diverse selection of vision and language instruction-following datasets in our study, including LLaVA [8], Mini-GPT4 [18], A-OKVQA [14], COCO Caption [7], and OCR VQA [10]. These datasets encompass a wide array of applications and domains, thereby facilitating the comprehensive evolving of our model's performance.

In order to present the text in a consistent, instruction-following format, we adopt the prompt delineated in Table 2 as a template for structuring these datasets. By adhering to a standardized format, we ensure that our model is better equipped to process the information and respond accordingly.

It is important to note that the COCO Caption dataset generally does not include instructional content, as it predominantly consists of descriptive captions. To overcome this limitation and incorporate instructional data, we employ the GPT-4 [11] model to generate pertinent instructions for the COCO Caption dataset. This integration of synthesized instructions enriches the dataset, enabling our model to achieve a more robust capabilities in processing and responding to human instructions.

Table 3 showcases a variety of examples illustrating the instructions generated for the COCO Caption dataset, demonstrating the effectiveness of our approach in adapting the dataset to better suit our research objectives.

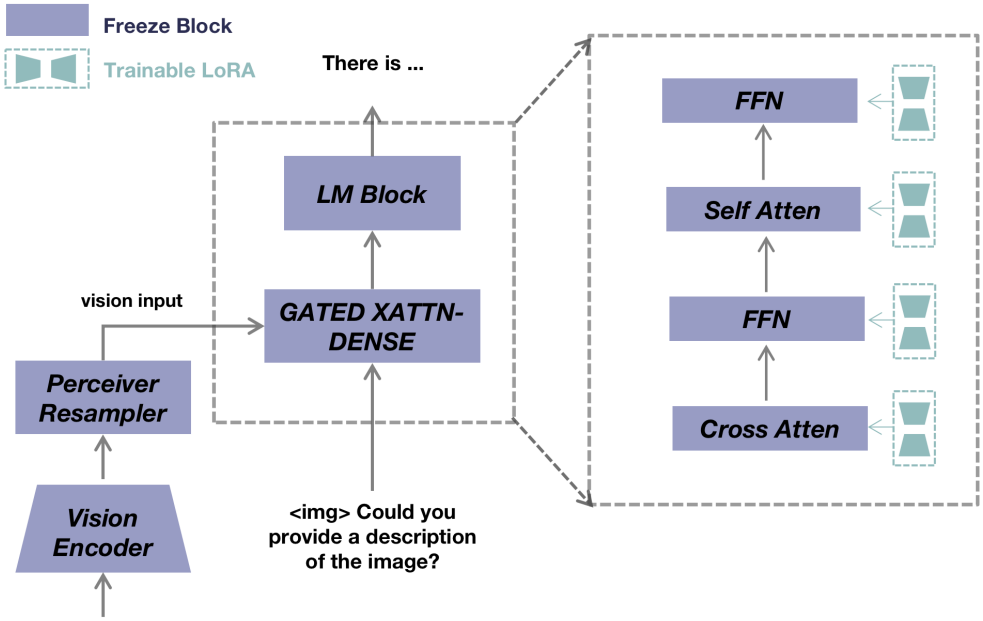

Figure 1: The overall framework of MultiModal-GPT. MultiModal-GPT consists of a vision encoder, a perceiver resampler to receive the spatial features from the vision encoder, and a language decoder which is conditioned on the spatial features from the perceiver resampler by cross-attention in order to encode the feature of vision into text. We freeze the whole open-flamingo model and add LoRA to the self-attention part, the cross-attention part, and the FFN part in the language decoder to finetune MultiModal-GPT.

3 Method

3.1 Architecture

The proposed MultiModal-GPT is based on the open-flamingo model [1]. As shown in Figure 1, MultiModal-GPT consists of a vision encoder from CLIP [13], a perceiver resampler to receive the spatial features from the vision encoder, and a language decoder LLaMA [16]. Note that the language decoder is conditioned on the spatial features from the perceiver resampler by cross-attention in order to encode the feature of vision into text. Please refer to [1] for more details of the model architecture.

3.2 Joint Training

We use both language-only instruction-following data and vision and language instruction-following data to train the MultiModal-GPT jointly. As shown in Fig.1, We freeze the whole open-flamingo model and add LoRA [4] to the self-attention, cross-attention, and FFW part in the language decoder to finetune MultiModal-GPT. The MultiModal-GPT is trained by predicting the next token of the text, and only the {response} and <EOS> tokens in the input sequence are involved in the loss calculation.

4 Experiments

4.1 Implementation Details

We jointly train the MultiModal-GPT model using a comprehensive mix of language data and vision and language data sources to enhance its performance. The language datasets include Dolly 15k and Alpaca GPT4 [12], while the vision and language datasets encompass LLaVA [8], Mini-GPT4 [18], A-OKVQA [14], COCO Caption [7], and OCR VQA [10]. Other vision and language instruction datasets, such as MultiInstruct [17] can also be explored, and we left it as future work. This

combination of datasets aims to provide a diverse and rich training environment for the MultiModal-GPT model.

To effectively train the model, we incorporate the entire text corpus from the Dolly 15k and Alpaca GPT4 datasets. Similarly, we include all image-text pairs available from the LLaVA and Mini-GPT4 datasets to ensure adequate exposure to various contexts and situations. However, the quality of the A-OKVQA, COCO Caption, and OCR VQA datasets is considered inferior compared to LLaVA and Mini-GPT4. To account for this disparity while still benefiting from the additional data, we include a random sample of 5000 image-text pairs from the A-OKVQA dataset and 512 image-text pairs each from the COCO Caption and OCR VQA datasets in the training process.

To train the model, we utilize 8 A100 GPUs and complete the training process within a single epoch. The batch size per GPU for both vision and language instruction following data and language-only instruction following data is set to 1. We use gradient accumulation, and update the parameters of the LoRA every 16 iterations. Each iteration encompasses one vision-language pair and one language-only instruction data. Consequently, the aggregate batch size amounts to 256. We employ a learning rate of 1e-5, using a cosine learning rate scheduler to adjust the learning rate during the training process.

4.2 The Quality of Data Matters

In order to enhance the diversity of vision and language data, we initially incorporated a multitude of datasets in our preliminary experiments. However, we observed that the quality of certain datasets, including VQA v2.0 [3], OKVQA [9], GQA [5], CLEVR [6], and NLVR [15], was suboptimal. This is primarily due to the fact that the {response} in these datasets is restricted to one or two words (e.g., yes/no). Consequently, when these datasets are incorporated into the training process, the model exhibits a tendency to generate answers comprising merely one or two words. This brevity is not conducive to user-friendliness. As a result, we have opted to exclude these datasets from the final version of our study.

4.3 Demos

We present a variety of demos that exhibit the capabilities of MultiModal-GPT in engaging in conversation with humans. As illustrated in Figure 2, MultiModal-GPT can successfully identify lasagna and provide a comprehensive recipe for preparing the dish when prompted by the user in the initial dialogue. In the subsequent dialogue, MultiModal-GPT offers reasonable suggestions on where to dine when the user inquires about potential eateries. This exemplifies the model's capacity for maintaining an ongoing dialogue.

Additionally, Figure 4 demonstrates MultiModal-GPT's counting abilities, while Figure 6 showcases its OCR capabilities. MultiModal-GPT is also adept at addressing general inquiries regarding users' travel plans, as depicted in Figure 7. More demos can be found at https://github.com/open-mmlab/Multimodal-GPT.

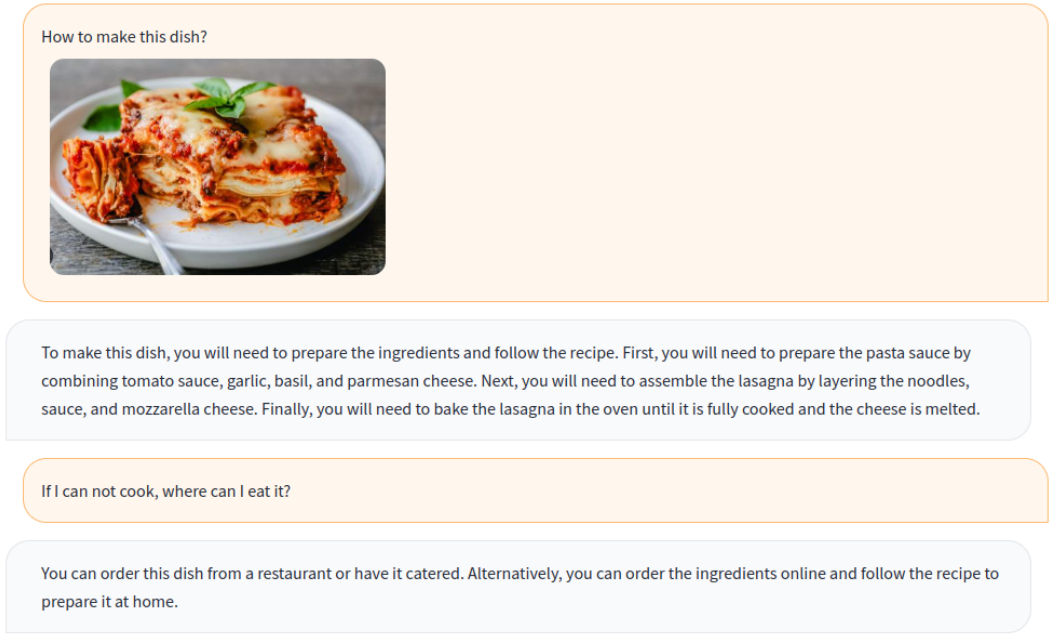

Figure 2: The MultiModal-GPT can give a recipe to bake lasagna, and tell users where to eat it.

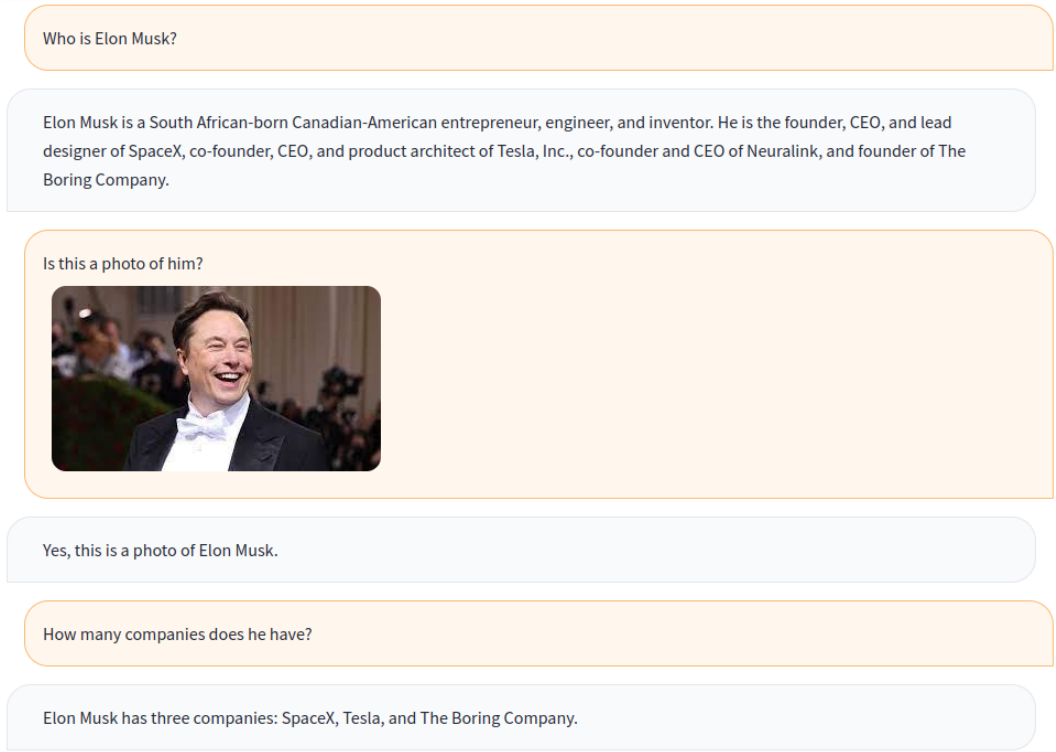

Figure 3: The MultiModal-GPT can recognize the image of Elon Musk, and answer questions about Elon Musk.

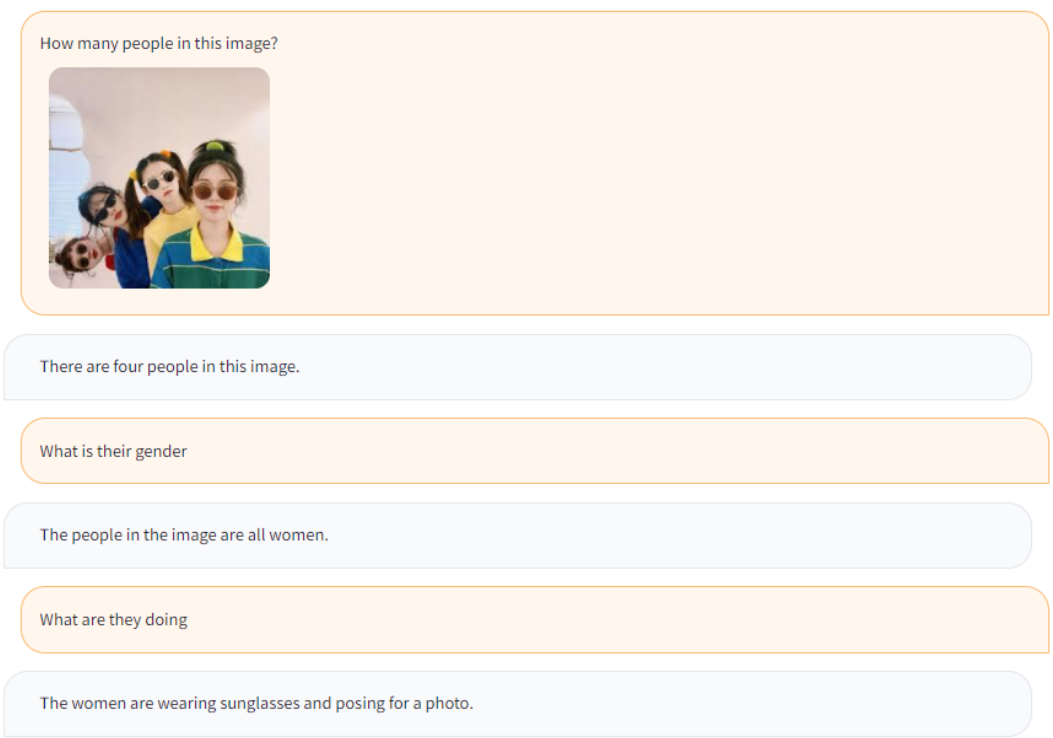

Figure 4: The MultiModal-GPT knows there are 4 women in the image, and knows what they are doing.

Figure 5: The MultiModal-GPT knows the film and knows which studio made the film.

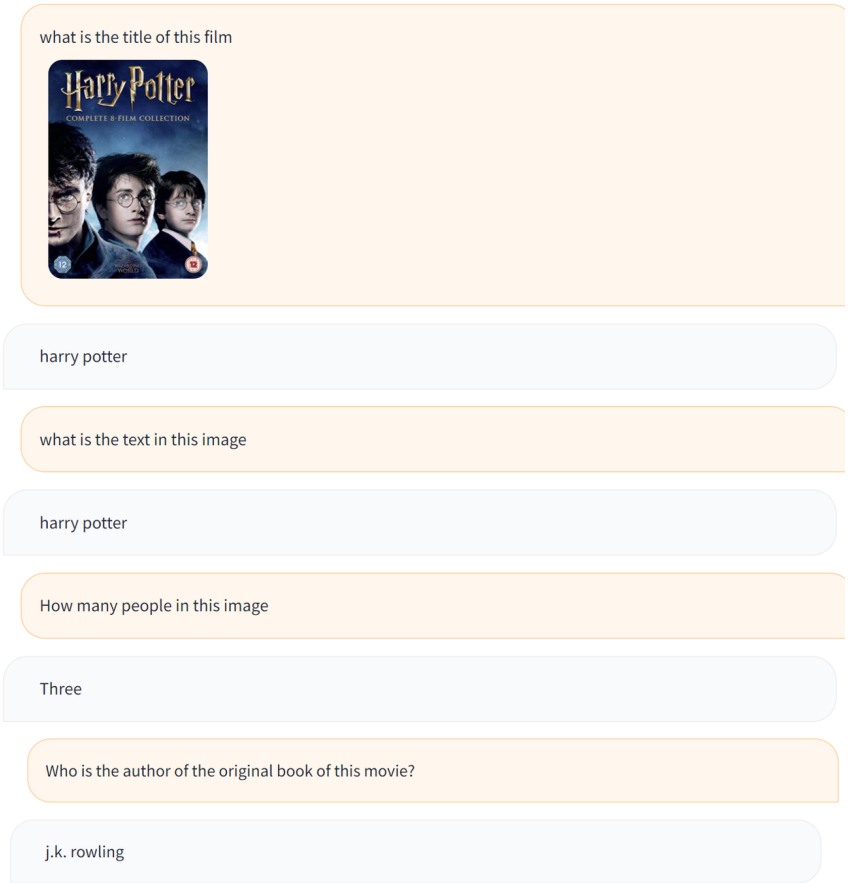

Figure 6: The MultiModal-GPT can recognize the characters in the image and knows the author of the book.

What can I do if I travel there?

If you travel to the Hawaiian island of Kauai, you can enjoy a variety of activities such as swimming, snorkeling, surfing, hiking, and sightseeing. You can explore the island's natural beauty by visiting the Waimea Canyon, also known as the "Grand Canyon of the Pacific", and the Na Pali Coast, which boasts breathtaking cliffs and waterfalls. You can also explore the island's cultural heritage by visiting the National Tropical Botanical Garden and the Kauai Museum. Additionally, the island is home to several resorts and hotels, where you can stay and enjoy amenities such as swimming pools, spas, and restaurants.

How much will it cost?

The cost of traveling to Kauai can vary depending on your choice of accommodation, transportation, and activities. You can find affordable accommodation options, such as hostels and Airbnb rentals, as well as more luxurious options, such as hotels and resorts. Transportation costs can vary depending on your mode of transportation. For example, if you choose to rent a car, you will have to pay for gas and parking fees. If you prefer to use public transportation, you will have to pay for bus tickets. The cost of activities, such as swimming, snorkeling, surfing, hiking, and sightseeing, can also vary depending on your preferences and the availability of equipment.

What else should I prepare for this trip?

To prepare for your trip to Kauai, you should research the island's attractions, activities, and transportation options. You should also check the weather forecast to ensure that you are prepared for any potential weather conditions. Additionally, you may want to bring sunscreen, a swimsuit, and waterproof clothing, depending on your planned activities and the weather.

Figure 7: The MultiModal-GPT can answer general questions about the traveling of users.

Please give a detailed description of this image

The image shows a beautiful lake surrounded by lush green grass and trees. The lake is filled with water lilies, creating a tranquil and serene atmosphere.

The lake is situated in a mountainous landscape, with a mountain peak visible in the background.

There are several birds flying around the lake, adding to the tranquility and natural beauty of the scene.

What is the season in the picture

The season in the image is summer, as evidenced by the lush green grass and the presence of water lilies in the lake.

If I want to travel to some places like in this image, what is your suggestion

If you want to experience a tranquil and serene environment similar to the one in the image, you could visit a natural lake surrounded by lush greenery and mountains.

Figure 8: The MultiModal-GPT can generate a detailed description for the image and has the ability to reason the season in the image.

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